Prodigy Anesthesia Quick Review

Brachial Plexus Block

How do you anesthetize the musculocutaneous nerve?	By injecting 3-5 mL of local anesthetic into the coracobrachialis muscle. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1112.*
Which nerve roots form the superior, middle, and inferior trunks of the brachial plexus?	The C5 and C6 nerve roots form the superior trunk, C7 forms the middle trunk, and C8 and T1 form the inferior trunk. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1108.*
What nerve roots contribute to the radial nerve?	C5 to T1 *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1108.*
What nerve roots contribute to the ulnar nerve?	C8-T1 *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1108.*
Which nerve roots contribute to the median nerve?	C6-T1 *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1108.*
For what procedures is an axillary brachial plexus block best suited?	An axillary brachial plexus block is best suited for procedures at or below the elbow. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1111.*
The brachial plexus is formed from the rami of which nerves?	It is formed from the rami of C5-T1. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1108.*
What three nerves may need to be blocked separately to augment an axillary block?	The musculocutaneous, medial brachial cutaneous, and intercostobrachial nerves may need to be blocked separately as they exit the brachial plexus sheath high in the axilla. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1112.*
How does an interscalene brachial plexus block affect respiratory function and why?	An interscalene brachial plexus block with 40-50 mL of local anesthetic ensures a nearly 100% incidence of ipsilateral hemidiaphragmatic paresis. Reducing the volume of local anesthetic to 30 mL, however, does not reduce the incidence of paresis and its pulmonary effects significantly. The paresis is typically complete within 15 minutes of injection and normal hemidiaphragmatic motion typically returns in about 3-4 hours. *Yao FF. Anesthesiology: Problem-Oriented Patient Management. 7th ed. Philadelphia: Lippincott, Williams, and Wilkins; 2012: 1035.*

Copyright 2015 Prodigy Anesthesia, Inc. Unlawful to duplicate, distribute, or sell this document.

Prodigy Anesthesia Quick Review

Brachial Plexus Block

For what procedures is an interscalene brachial plexus block sufficient?	A brachial plexus block is usually indicated for procedures on the shoulder, upper arm and forearm. It is usually not sufficient for procedures on the hand, however. The C8 and T1 nerves which innervate the ulnar aspect of the forearm are frequently missed as well. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 959.*
The presence of cough or chest pain while performing a supraclavicular, infraclavicular, or interscalene brachial plexus block may indicate what?	Chest pain or cough during the procedure can be an indication of an inadvertant pneumothorax. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 961.*
What are the most serious potential complications of an interscalene brachial plexus block?	Vertebral artery injection, high epidural or spinal anesthesia, and pneumothorax. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 961.*
What is Horner's syndrome and how does it relate to interscalene brachial plexus blocks? What other nerves are often blocked incidentally when performing an interscalene block?	Horner's syndrome occurs as a result of incidental blockade of the stellate ganglion and causes anhidrosis, myosis, and ptosis. Incidental blockade of the phrenic nerve will result in ipsilateral hemidiaphragmatic paralysis, which can be significant in patients with severe respiratory problems. Incidental blockade of the recurrent laryngeal nerve will result in hoarseness from unilateral paralysis of the vocal cord. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 988.*
What are the major potential complications of a supraclavicular brachial plexus block?	Pneumothorax, hemothorax, and chylothorax are potential complications of a supraclavicular brachial plexus block. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 988.*

Copyright 2015 Prodigy Anesthesia, Inc. Unlawful to duplicate, distribute, or sell this document.

Brachial Plexus Block

How is a supraclavicular brachial plexus block performed?	Just posterior to the subclavian artery between the clavicle and the first rib, the brachial plexus is tightly compacted. Because of this, a block performed here will achieve adequate anesthesia of the entire arm and hand. The patient is positioned supine with the head turned to the opposite side. A skin wheal is raised posterior to the subclavian artery at the most inferior point of the interscalene groove. A 22-gauge, 1.5 inch needle enters the wheal posterior to the pulse and is directed caudally until a paresthesia or muscle contraction (if using a nerve stimulator) is elicited. If blood is aspirated, redirect the needle posteriorly and laterally. If the first rib is encountered, walk the needle in an anterior to posterior manner. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 986-988.*
During an axillary brachial plexus block, you can choose to begin the injection high in the axilla, in the middle, or near the inferior border of the axilla. What is the benefit of choosing a higher approach?	Any one of these locations can be used for an axillary brachial plexus block, but the higher the approach, the greater the chances that the block will anesthetize the musculocutaneous nerve before it exits the sheath. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1111.*

Copyright 2015 Prodigy Anesthesia, Inc. Unlawful to duplicate, distribute, or sell this document.